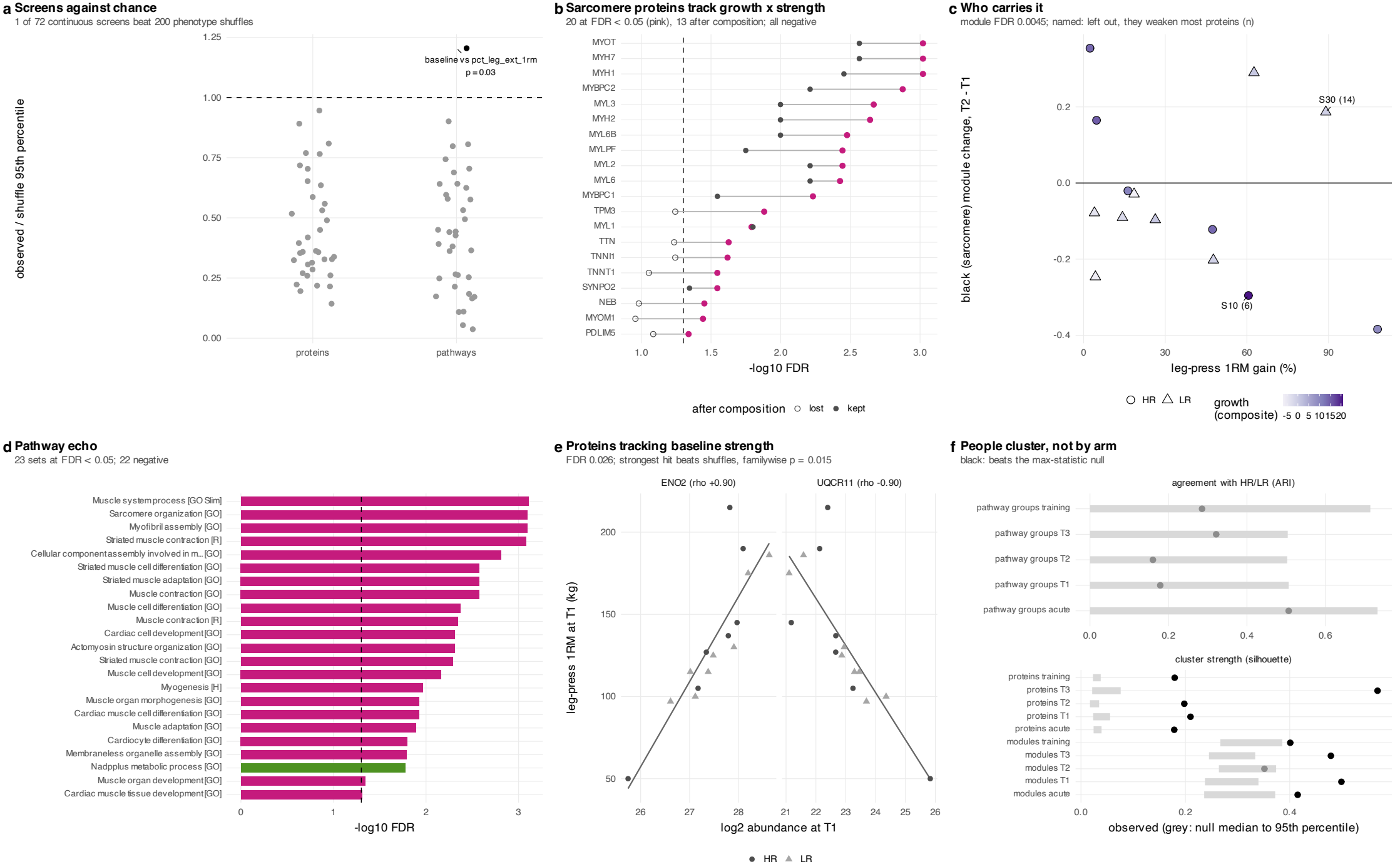
a Screens against chance
1 of 72 continuous screens beat 200 phenotype shuffles
observed / shuffle 95th percentile
baseline vs pct_leg_ext_1rm
p = 0.03
proteins
pathways
b Sarcomere proteins track growth x strength
20 at FDR < 0.05 (pink), 13 after composition; all negative
MYOT
MYH7
MYH1
MYBPC2
MYL3
MYH2
MYL6B
MYLPF
MYL2
MYL6
MYBPC1
TPM3
MYL1
TTN
TNNI1
TNNT1
SYNPO2
NEB
MYOM1
PDLIM5
-log10 FDR
after composition lost kept
c Who carries it
module FDR 0.0045; named: left out, they weaken most proteins (n)
black (sarcomere) module change, T2 - T1
S30 (14)
S10 (6)
leg-press 1RM gain (%)
HR LR
growth (composite)
d Pathway echo
23 sets at FDR < 0.05; 22 negative
Muscle system process [GO Slim]
Sarcomere organization [GO]
Myofibril assembly [GO]
Striated muscle contraction [R]
Cellular component assembly involved in m... [GO]
Striated muscle cell differentiation [GO]
Striated muscle adaptation [GO]
Muscle contraction [GO]
Muscle cell differentiation [GO]
Muscle contraction [R]
Cardiac cell development [GO]
Actomyosin structure organization [GO]
Striated muscle contraction [GO]
Muscle cell development [GO]
Myogenesis [H]
Muscle organ morphogenesis [GO]
Cardiac muscle cell differentiation [GO]
Muscle adaptation [GO]
Cardiocyte differentiation [GO]
Membraneless organelle assembly [GO]
Nadpplus metabolic process [GO]
Muscle organ development [GO]
Cardiac muscle tissue development [GO]
-log10 FDR
e Proteins tracking baseline strength
FDR 0.026; strongest hit beats shuffles, familywise p = 0.015
ENO2 (rho +0.90)
UQCR11 (rho -0.90)
leg-press 1RM at T1 (kg)
log2 abundance at T1
HR LR
f People cluster, not by arm
black: beats the max-statistic null
agreement with HR/LR (ARI)
pathway groups training
pathway groups T3
pathway groups T2
pathway groups T1
pathway groups acute
cluster strength (silhouette)
proteins training
proteins T3
proteins T2
proteins T1
proteins acute
modules training
modules T3
modules T2
modules T1
modules acute
observed (grey: null median to 95th percentile)